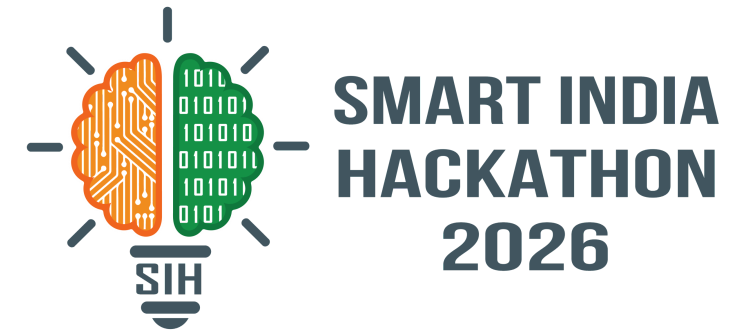


SMART INDIA HACKATHON 2026



- **Problem Statement ID – SIH 26017**
- **Problem Statement Title- Predictive Analytics System for Early Detection of Land Acquisition Delays**
- **Theme- Smart Automation**
- **PS Category- Software**
- **Team ID-135113**
- **Team Name-Terminal Terminators**



Predict.Explain.Mitigate

AI-driven early warning, legal-timeline forecasting , actionable intervention and prevention planning for in land acquisition.



HTML
 Tier 3 – Frontend / User App
HTML / CSS / JavaScript
React chart components
Streamlit

```

graph LR
    PI[Project Inputs] --> LFP[Leak-Free Preprocessing]
    LFP --> DP[DELAY PREDICTION]
    LFP --> SF[SURVIVAL FORECAST]
    
    subgraph DP_Box [DELAY PREDICTION]
        XGB[XGBoost] --> LRM[Logistic Regression Meta-Learner]
        LGB[LightGBM] --> LRM
        CB[CatBoost] --> LRM
        ET[ExtraTrees] --> LRM
    end
    
    subgraph SF_Box [SURVIVAL FORECAST]
        RSF[Random Survival Forest] --> SF_S[S(t)]
        DS[DeepSurv] --> SF_S
    end
    
    LRM --> PC[Platt Calibration]
    PC --> DPO[Delay Probability + Delay Days]
    
    SF_S --> DRC[90 / 180 / 270 / 365 / 540 / 730 Day Risk Curve]
    
    DPO --> EC[EXPLAINABILITY CHAIN]
    DRC --> EC
    
    subgraph EC_Box [EXPLAINABILITY CHAIN]
        TS[TreeSHAP] --> RD[Risk Drivers]
        RD --> CPR[Critical Path + ROI]
        CPR --> RI[Recommended Intervention]
    end
    
    EC_Box --> CL[CONTINUOUS LEARNING]
    
    subgraph CL_Box [CONTINUOUS LEARNING]
        SO[Saved Outcomes] -.-> PSD[PSI Drift Detection]
        PSD -.-> R[Retraining]
        R -.-> TVG[Triple Validation Gate]
        TVG -.-> VM[Versioned Model]
        VM -.-> P[Production]
        P -.-> SO
    end

```

Project Outcome Prediction

Project Inputs → **Leak-Free Preprocessing**

DELAY PREDICTION

- XGBoost
- LightGBM
- CatBoost
- ExtraTrees

Logistic Regression Meta-Learner → Platt Calibration → Delay Probability + Delay Days

SURVIVAL FORECAST

- Random Survival Forest
- DeepSurv

Survival Function $S(t)$ → 90 / 180 / 270 / 365 / 540 / 730 Day Risk Curve

EXPLAINABILITY CHAIN

- TreeSHAP
- Risk Drivers
- Critical Path + ROI
- Recommended Intervention

CONTINUOUS LEARNING

- Saved Outcomes
- PSI Drift Detection
- Retraining
- Triple Validation Gate
- Versioned Model
- Production

MODEL QUALITY

R² ≥ 0.946	MAE ≈ 31.6 days	ECE ≤ 0.015	Uno's C-Index = 0.9028
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UNCERTAINTY

96% Conformal Coverage	Delay Interval Output Example: 280–342 days
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MLOPS / SYSTEM BENCHMARKS

Preprocessing: ~0.26 s	Base Models + Meta + Platt: ~308 s
Survival Engine: ~57 s	TreeSHAP Rebuild: ~22 s
Validation Gate: ~2.08 s	ONNX + Versioning ~24.7 s

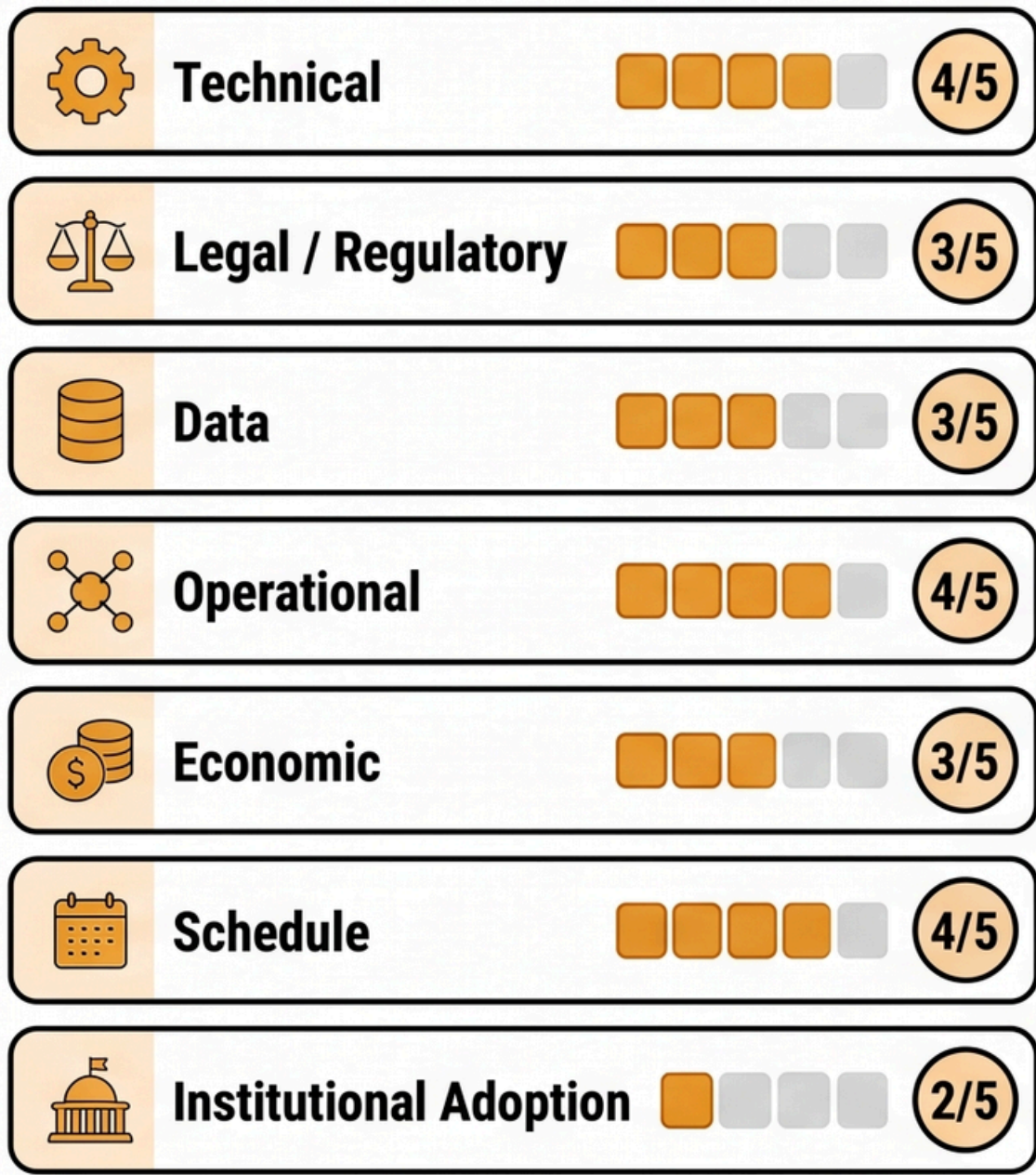
R²-Regression Accuracy
MAE — Mean Absolute Error
ECE — Expected Calibration Error

05 – Simulate + Govern

What-if analysis + drift + retraining

FEASIBILITY AND VIABILITY

FEASIBILITY SCORECARD

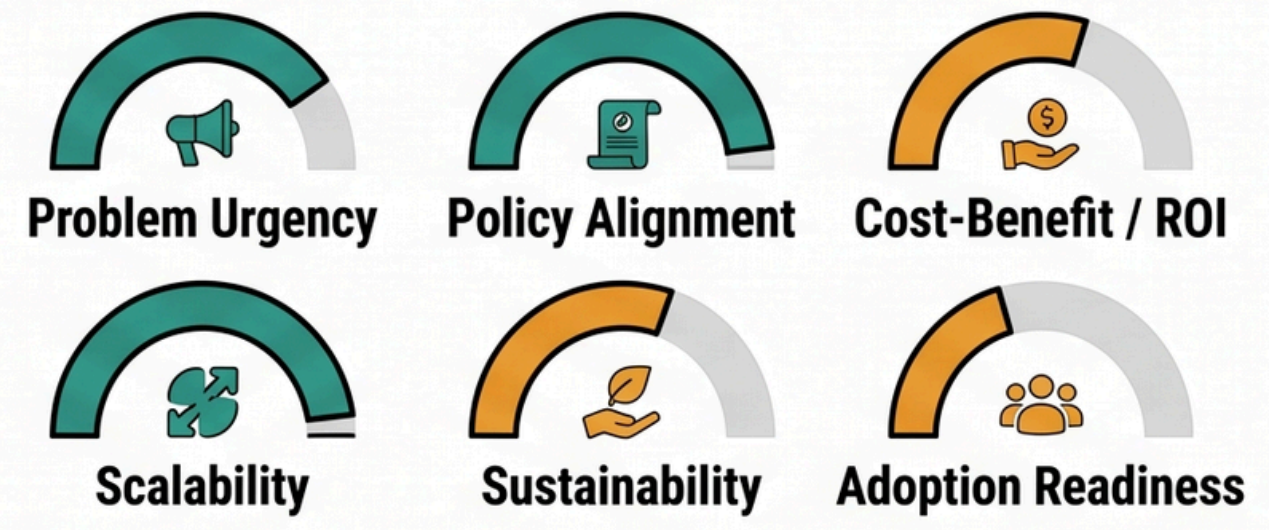


Overall Feasibility: Medium



VIABILITY & RISK OVERVIEW

Viability Dashboard



Viable, conditional on data partnerships, legal review, and pilot adoption.

Key Risks

Likelihood	Impact		
	Low	Medium	High
	High	R4 R10	R1
	Medium	R7 R3 R6	R2 R5 R8
Low			R9

- R1 Data scarcity
- R2 Legal misread
- R4 Adoption resistance
- R5 API/security
- R6 GIS dependency
- R9 Regulatory change
- R7 Compute cost
- R8 Bias/fairness
- R10 Legacy integration

READINESS SIGNALS

Legal Anchor

RFCTLARR Act 2013 – Sec 11, 15, 19, 23/24, 38

Section 19: 365-Day Deadline

Governance Proof

- PSI Drift Monitoring
- 3-Gate Retraining
- 3-Version Rollback
- API Key / JWT
- Rate Limits
- Prompt-Injection Defense

Project Proof

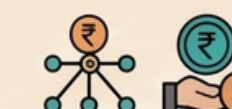
- Stacking Ensemble
- Conformal Prediction
- Survival Analysis
- SHAP + TabNet
- Critical-Path Engine
- /simulate Endpoint
- LA-7 Parser
- Docker Deployed

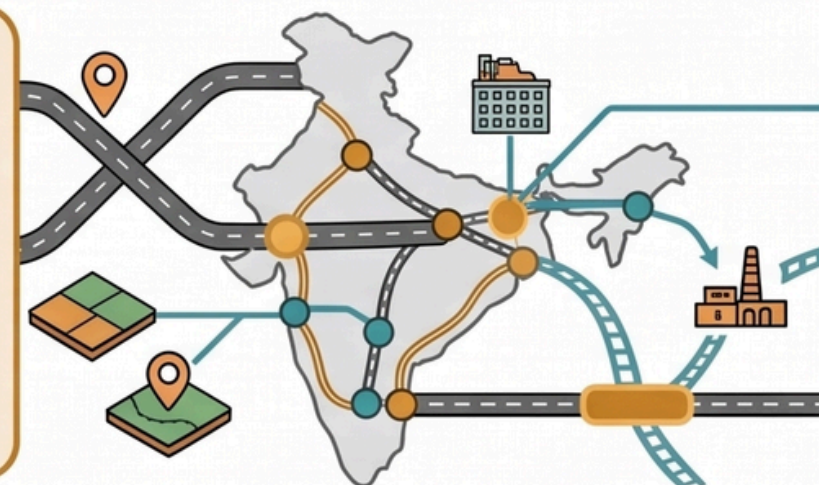
Feasibility: Medium. Viability: Conditional.
Next step: Pilot with a state department + legal review.

PROBLEM SCALE, STAKEHOLDERS & NATIONAL IMPACT

PROBLEM SCALE

 **70%+**
of infrastructure-project
time/cost overruns

₹100+  **Lakh Crore**
infrastructure investment /
exposure at national scale



COMMUNITY & IMPACT MANAGEMENT



Community and Local Stakeholders

Need: Public project timeline visibility +
compensation tracking + R&R action clarity

Problem-scale figures cited in project documentation, verify source methodology before final submission.

WHO NEEDS THE SYSTEM



Project Leadership

Project Directors / Infrastructure Agencies

Need: Early warning + schedule visibility + Intervention ROI



Ground Execution

District Land Acquisition Officers

Need: Statutory tracking + bottleneck diagnosis + action prioritization

**ONE SHARED
RISK VIEW**

Governance & Oversight

Government / Auditors / Enterprise Partners
Need: Explainable evidence + model health + reporting



NATIONAL IMPACT

CARD 1

ECONOMIC



-  Detect delay earlier
-  Prioritize highest-ROI intervention
-  Reduce avoidable schedule exposure

CARD 2

SOCIAL



-  Identify affected projects earlier
-  Support timely compensation / R&R
-  Improve officer prioritization

CARD 3

ENVIRONMENTAL



-  Surface clearance bottlenecks
-  Flag terrain / forest sensitivity
-  Support earlier corrective action

CARD 4

DIGITAL / GOVERNANCE



-  Standardize project risk assessment
-  Enable explainable AI decisions
-  Maintain model-health visibility

DATA → EARLY WARNING → TARGETED ACTION → FEWER UNCONTIOLLED DELAYS

Predict.Explain.Mitigate



Academic Research

- [1] **Breiman, L.** (2001) – Random Forests. Machine Learning. DOI: 10.1023/A:1010933404324
- [2] **Ishwaran, H.** et al. (2008) – Random Survival Forests. Annals of Applied Statistics. DOI: 10.1214/08-AOAS169
- [3] **Chen, T. & Guestrin, C.** (2016) – XGBoost: A Scalable Tree Boosting System. KDD. DOI: 10.1145/2939672.2939785
- [4] **Katzman, J.** et al. (2018) – DeepSurv: Personalized Treatment Recommender System Using a Cox Proportional Hazards Deep Neural Network. BMC Medical Research Methodology. DOI: 10.1186/s12874-018-0482-1



Standards and Regulations

- [1] **Government of India**, 2013 – **RFCTLARR Act, 2013** – Land acquisition statutory workflow
- [2] **Government of India**, 2023 – **Digital Personal Data Protection Act, 2023** – Digital personal-data governance
- [3] **Government of India**, 2022 – **National Geospatial Policy 2022** – Geospatial data and interoperability context
- [4] **ISO**, 2018 – ISO 31000.2018 Risk Management – Guidelines – Risk identification and treatment



Open Source and Technical Docs

- [1] **scikit-learn 1.9.x** – scikit-learn.org/stable
- [2] **XGBoost 3.4.x** – xgboost.readthedocs.io
- [3] **FastAPI** – fastapi.tiangolo.com
- [4] **MLflow 2.9+** – mlflow.org/docs

 **Project repository:** <https://github.com/mai-lakshya/Slh>  **Video Link:** <https://youtu.be/udnzfxzeRuo>
 **Simulation/Demo Link:** <https://drive.google.com/drive/folders/1mcRIA34NsPGypRmftliZy71pt720I7e1?usp=sharing>